Part One Concept Reference

Central concepts and notation for Chapters 1–14

Jinsong Chen

Table of contents

Chapter 1: The Quantitative Research Journey . . . 2
Chapter 2: Data, Distributions, and Variation . . . 3
Chapter 3: Estimates, Uncertainty, and Statistical Tests . . . 6
Chapter 4: Group Comparisons with Analysis of Variance . . . 8
Chapter 5: Description and Basic Inference with R . . . 9
Chapter 6: Group Comparisons and Statistical Modelling with R . . . 10
Chapter 7: Simple Linear Regression . . . 10
Chapter 8: Multiple Regression I: Adjusted Relationships, Group Comparisons, and Model Comparison . . . 12
Chapter 9: Multiple Regression II: Model Assumptions, Diagnostics, and Respecification . . . 14
Chapter 10: Interaction and Moderation . . . 15
Chapter 11: Binary Logistic Regression . . . 17
Chapter 12: Extending the GLM: Nominal, Ordinal, and Count Outcomes . . . 19
Chapter 13: Prediction, Cross-Validation, and Regularization . . . 21
Chapter 14: Regression Pitfalls and Causal Design . . . 23
Argument Chapter A: How to Read a Quantitative Study as an Argument . . . 24
Argument Chapter B: Extending the Argument-Based Approach to Quantitative Studies . . . 25
Procedures and Their Conditions . . . 25
Shared Notation . . . 29

This reference gathers the Key terms tables of Chapters 1–14 and of Argument Chapters A and B, followed by the procedures of Chapters 3 and 4 and the notation that several chapters share. Each table appears under its chapter's heading and lists the concepts that chapter explains principally: each term with its common alternatives, and its meaning or the distinction that matters most. The reference is meant to be consulted while reading rather than read through: a term's first use, its worked example, and the reasoning around it stay in the chapter that teaches it. Chapters 1–9 are available on the website; the tables for Chapters 10–14 point forward to chapter readings that are not yet available.

A few conventions govern the rows. Names that pick out the same concept share one entry, so *mean* and *arithmetic mean* appear together, and *outcome*, *response*, and *dependent variable* are explained as one analytical role. A more specific role, such as treatment, exposure, or

confounder, carries additional design or substantive meaning that the study itself must supply. Related concepts with different meanings, such as mean and median or SD and SE, keep separate entries. A term is listed under the chapter that explains it principally rather than the chapter that first mentions it, so a brief early introduction and a later full account can coexist: Chapter 1 introduces cases and samples through its studies, while Chapter 2 develops their use in reading records and calculations. Argument Chapters A and B, *How to Read a Quantitative Study as an Argument* and *Extending the Argument-Based Approach to Quantitative Studies*, extend the reading of one question-level argument to several statistical strands and coordinated questions; the notation table abbreviates them as *Arg. A* and *Arg. B*.

Chapter 1: The Quantitative Research Journey

Table 1
Key terms of Chapter 1.

Term and common alternatives	Meaning
Question-answer pair	What a study asks and the answer it eventually offers for that question. Each separately owed substantive answer has its own argument.
Traceable information	Specific study content through which an argument component's function can be identified at a documentary location. Identifying that contribution precedes judging its adequacy.
Estimand	The target quantity sought for the named units, conditions, outcome, and scope. It can concern a fixed dataset, a population, a comparison, or prediction performance.
Outcome variable (response variable; dependent variable)	The variable an analysis seeks to describe, compare, or predict. These alternative names share that role in many analyses; a variable's role depends on the question.
Predictor variable (explanatory variable; independent variable)	A variable used to represent an outcome's relationship with other information. These analytical-role labels alone establish neither a causal effect nor statistical independence.
Sampling (recruitment; selection into a study)	How units come to enter a study, by probability selection, volunteering, or availability. It is contrasted with assignment, which concerns the conditions allocated to participating units.
Assignment (allocation)	The allocation of study conditions to participating units, by any mechanism. Actual treatment received may differ from the assigned condition.
Random sampling (probability sampling)	Selecting units from a defined population or sampling frame through a probability mechanism that gives every eligible unit a known, nonzero chance of selection, not necessarily the same chance. "Random selection" alone can describe other chance-based decisions.
Random assignment (random allocation)	Allocating study conditions to participating units through a specified probability mechanism. The assigned units may be individuals or groups; realized groups need not be identical.

Term and common alternatives	Meaning
Causal effect (counterfactual contrast)	The difference between the outcomes that would occur for the same units or target population under two specified conditions, only one of which is observed for each unit. The contrast may be zero; interpreting a comparison causally needs design and identification support.
Confounding	Mixing an exposure's effect with effects of other factors that influence both exposure and outcome. Such a factor is a confounder, a specific causal role.
External validity (generalizability)	Support for extending findings to other people, settings, or conditions. The intended extension determines which similarities and differences matter.
Unit	The entity selected, assigned, measured, or analyzed in a particular operation. The units of assignment, measurement, and analysis can differ, and all are distinct from measurement units such as points or minutes.
Evidence	Observations and contextual records contributing to an answer, including information about how they were produced.
Justification	Reasons and conditions invoked to support a comparison, analytical move, or interpretation.
Inference	Reasoning and analysis connecting evidence and justification to a conclusion.
Bounded claim	The reported conclusion with its uncertainty, conditions, and scope: in the statistical layer, the conclusion about the estimand; in the substantive layer, the question-facing answer with its qualifications.

Chapter 2: Data, Distributions, and Variation

Table 2
Key terms of Chapter 2.

Term and common alternatives	Meaning
Case	The entity or measurement occasion described in a dataset. Depending on the study, a case may be a student, a class, or one measurement occasion for a student.
Variable	A characteristic that can take different values across cases or occasions. A dataset records values of one or more variables for its cases.
Quantitative variable (numerical variable)	A variable expressing an amount, including continuous measurements and discrete counts or scores. Its units and scale determine what numerical differences mean.
Categorical variable	A variable whose values identify categories. Nominal categories have no intrinsic order; ordinal categories have an order; a binary variable has two categories.

Term and common alternatives	Meaning
Population	The collection of units a question concerns. It may be the participating students themselves or a wider collection to which the findings might extend.
Sample	A selected set of units from a larger collection. Calling a set a sample does not by itself explain how it was selected or which population it represents.
Analysis sample	The units whose data contribute to a particular analysis. Eligibility, missing measurements, and exclusions can make this set and its denominator differ across analyses.
Sampling design (sampling scheme)	The specified procedure for selecting a sample, including selection stages, grouping, and, for probability sampling, the selection probabilities.
Simple random sampling	Selection in which every possible sample of the specified size has the same probability of being chosen.
Empirical distribution (observed distribution; data distribution)	How the observed values are distributed across values or intervals, including their frequencies, center, spread, and shape. A probability model is a separate description of possible values.
Skewness and kurtosis (left or negative skew; right or positive skew; tail weight)	Skewness describes asymmetry: in a right (positive) skew the long tail points toward larger values and the mean usually lies above the median; in a left (negative) skew the long tail points toward smaller values and the mean usually lies below the median. Kurtosis describes tail weight relative to a Normal distribution with the same mean and SD, not peakedness.
Mean (arithmetic mean)	The sum of the numerical values divided by their count. Every included value contributes, so unusual values can shift the mean.
Median	The middle value after ordering the data, or the average of the two middle values when the count is even. It describes the center by position.
Quartile	A cut point dividing ordered data into quarters. The first and third quartiles mark approximately the lower and upper edges of the middle half; calculation conventions can differ.
Interquartile range (IQR)	The third quartile minus the first quartile: the width of the middle half of the ordered data, expressed in the variable's original units.
Variance	A measure of spread based on squared deviations from the mean. The usual sample variance divides their sum by $n - 1$ and is expressed in squared units.
Standard deviation (SD)	The square root of the variance, expressing spread around the mean in the variable's original units. Applied to individual observations, it describes variation among their values; applied to a statistic's sampling distribution, it is the SE.
Grand mean	The mean across all included cases. It equals the average of the group means weighted by their group counts, so larger groups contribute more.
Degrees of freedom (df)	A count reflecting freely varying information after constraints or fitted quantities are taken into account. The n deviations from their own mean sum to zero, leaving $n - 1$ free.
Standardized value (z score)	A value minus a specified mean, divided by a specified standard deviation. It expresses distance in SD units; standardizing alone preserves the distribution's shape.

Term and common alternatives	Meaning
Normal distribution (Gaussian distribution)	A symmetric, bell-shaped probability model described by a mean and standard deviation. The standard Normal has mean zero and SD one; observed data need not follow this model.
Conditional proportion	The proportion in one category among cases satisfying a stated condition. Its denominator is the count in that conditioning group, such as students in one education-intention category.
Covariance	A summary based on products of paired deviations from the two means. Its sign shows the direction of joint variation; its magnitude depends on both variables' units and spreads.
Pearson's correlation (Pearson's r; product-moment correlation)	Covariance divided by the product of the two SDs. For variables with nonzero spread, it describes direction and strength of linear association on a scale from -1 to 1.
Sampling distribution	The distribution of a statistic, such as a sample mean, across repeated samples generated by a stated sampling procedure or model. Repeated sample summaries in a simulation approximate this distribution.
Sampling error	The part of an estimate's departure from the population quantity that arises from observing a sample rather than the whole population. It varies across repetitions, and its spread is the SE; coverage, nonresponse, and measurement error are separate.
Standard error (SE)	The standard deviation of a statistic's sampling distribution under a stated sampling procedure or model. It describes spread across repeated sample summaries, distinct from individual-value SD and one sample's difference from the population quantity.
Design effect (deff)	The factor by which a sampling design multiplies the variance of a statistic relative to simple random sampling of the same size. For a mean from equal clusters of m units it is about $1 + (m - 1)\,\mathrm{ICC}$, where the intraclass correlation (ICC) is the share of outcome variance lying between clusters: units in one cluster supply overlapping information.
Data matrix	A rectangular arrangement with cases in rows and variables in columns; x_{ij} denotes case i on variable j.
Binary variable (0/1 coding)	A binary variable has two possible categories; a 0/1 coding marks membership or occurrence. Its sample mean is the proportion coded 1.
Linear transformation	$Y = a + bX$ changes the center to $a + bE(X)$ and the SD to $\lvert b\rvert SD(X)$; a negative b also reverses order.
Bernoulli; binomial; categorical; Poisson; lognormal; logistic	Respectively a binary outcome, a count of successes in fixed independent equal-probability trials, one category, an event count under a specified rate model, a positive variable with Normal log, and a continuous symmetric distribution used here as a preview. Logistic regression later models Bernoulli outcomes.
Density; cumulative distribution function; quantile	Density determines probability by area for a continuous variable; the CDF gives $P(X \leq x)$; a quantile is a value reaching the stated cumulative probability.
Rank correlation; point-biserial correlation	Spearman uses ranks; Kendall summarizes concordance; point-biserial is Pearson correlation with one binary variable coded 0/1. Each addresses a specified relationship.

Term and common alternatives	Meaning
Partial correlation	Correlation between two variables after their linear relationships with stated controls are removed by regression; it describes residual association and does not establish causal adjustment.
Parameter	A numerical characteristic of a population or probability model, such as a population mean. A parameter may be the estimand; the question determines which quantity is sought.
Statistic	A quantity calculated from sample data, such as a sample mean or proportion. The same statistic can summarize the sample or serve as an estimate for a wider target.

Chapter 3: Estimates, Uncertainty, and Statistical Tests

Table 3
Key terms of Chapter 3.

Term and common alternatives	Meaning
Estimate	A numerical approximation to an estimand, calculated from the available data. A point estimate expresses the approximation as one value.
Estimator	The rule used to calculate an estimate from data. Its bias and precision concern how the rule behaves across the repetitions specified by the study's procedure or model.
Bias (of an estimator)	The difference between an estimator's expected value over the specified repetitions and the estimand. An unbiased estimator has bias zero; precision describes its spread, a separate quality.
Pooled variance (s_p^2; pooled within-group variance)	A weighted average of within-group sample variances, each weighted by its degrees of freedom: $s_p^2 = \{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2\}/(n_1 + n_0 - 2)$ for two groups. It estimates a variance the groups are assumed to share; its square root is the pooled SD.
Confidence interval (CI)	An interval expressing uncertainty about an estimand. Under its conditions, the procedure covers that estimand at its stated rate across repetitions; the rate describes the procedure.
Critical value	The cut-point of a reference distribution, such as the Normal or t, selected by a confidence or significance level. It multiplies the SE in an interval and marks the rejection region of a test.
Coverage	The proportion of repetitions in which an interval procedure contains the estimand under its conditions. The confidence level is the procedure's target coverage, not a probability for one realized interval.
Null hypothesis	The statistical proposition used as the reference for a test, such as equality of specified population means. The alternative hypothesis states the departure being considered.

Term and common alternatives	Meaning
Test statistic	A quantity calculated from data to assess departure from a null hypothesis. The test compares it with a reference distribution justified under that null and stated conditions.
p-value	Under a specified null hypothesis and conditions, the probability of a test result at least as extreme as the observed one, using the test's stated tail rule.
Significance level (alpha)	The threshold used for rejecting a null hypothesis, chosen to control the probability of rejecting a true null under the test's conditions. This error is called a Type I error.
One-sample z test; one-sample t test	Both compare a sample mean with a stated null mean using its SE; z uses a known process SD under its model, whereas t estimates that SD and uses the appropriate t reference. Neither choice follows an automatic sample-size cutoff.
Paired-samples t test	A one-sample t test of within-pair differences against a specified mean difference, with independent pairs under the stated model.
Equal-variance two-sample t test (pooled t test; Student's two-sample t test)	One of the two independent-samples t tests. It compares two independent group means under the condition that the populations share one variance: its SE is $s_p\sqrt{1/n_1 + 1/n_0}$, and its reference is t with $n_1 + n_0 - 2$ df.
Welch's unequal-variance t test (Welch test)	The other independent-samples t test. It compares two independent group means without assuming a common variance: its SE adds the two estimated mean variances, and its approximate t reference uses Welch–Satterthwaite df. With equal group sizes its statistic equals the equal-variance statistic, and only the df can differ.
Expected count	The count a cell of a two-way table would have under a stated null pattern such as independence, given the observed margins: row total times column total, divided by the overall total.
Pearson's chi-square statistic (X^2)	The sum over cells of squared differences between observed and expected counts, each divided by its expected count, compared with a chi-square reference with $(R-1)(C-1)$ degrees of freedom under the stated model.
Multiway table (three-way table; layered two-way tables)	A table that cross-classifies three or more categorical variables: a two-way table holds two variables, a three-way table three, and a four-way table four. A three-way table is read as two-way tables within the layers of the third variable; a chi-square test of one two-way table cannot test the variables jointly, which needs a method that uses all three variables at once, such as a model for multiway tables.
Standardized mean difference	A difference in means divided by a specified SD. The chosen denominator determines the standardized quantity; the raw difference retains the outcome's original units.
Power (statistical power)	The probability of rejecting the null under a specified alternative, design, and test. Failure to reject under that alternative is a Type II error; power is one minus its probability.

Chapter 4: Group Comparisons with Analysis of Variance

Table 4
Key terms of Chapter 4.

Term and common alternatives	Meaning
Factor (grouping factor)	In ANOVA, a categorical variable defining groups or conditions; its categories are levels. This meaning differs from a latent factor used to model shared indicator variation.
Analysis of variance (ANOVA)	A method for studying mean differences through sources of variation. One-way ANOVA uses one grouping factor, and multi-way (factorial) ANOVA uses two or more crossed factors, as two-way and three-way ANOVA do; each factor can have several levels.
Fitted value	The value a fitted model assigns to an observed case. In a one-way group-mean model, it is that case's group mean.
Residual	An observed outcome minus its fitted value. A residual is calculated from the fitted model; an error is the corresponding unknown departure in the population model.
Mean square (MS)	A sum of squares divided by its degrees of freedom. In classical ANOVA, the within-group mean square estimates the common error variance under the model's conditions.
F statistic (F ratio)	In classical ANOVA, a ratio of the mean square for a tested source to the appropriate error mean square. Its null F reference depends on the model and both degrees of freedom.
Omnibus test	A test of one joint null hypothesis about a whole pattern, such as equality of all group means. Rejecting it does not identify which comparisons differ or how large they are.
Eta squared	The proportion of total observed variation attributed to a specified source in an ANOVA decomposition. In one-way ANOVA, it is between-group divided by total sum of squares.
Contrast (linear contrast)	A weighted combination of means whose weights sum to zero, expressing a specific comparison. The means, weights, and target population determine the quantity compared.
Familywise error rate (FWER)	The probability of at least one Type I error among a defined family of tests. Procedures such as Bonferroni allocate the error allowance across that family.
Simultaneous confidence intervals	A set of intervals calibrated so that, under their conditions, all of them cover their estimands together at the stated familywise rate; Bonferroni, Tukey, and Dunnett are different calibrations for different families.
Cell mean and marginal mean	In a two-factor layout, a cell mean is the mean of the cases sharing one level of each factor; a marginal mean averages the cell means across the levels of the other factor, with stated weights.
Main effect	A comparison across one factor's levels averaged over the other factor's levels, using stated weights. It remains an average when the conditional pattern differs across levels.

Term and common alternatives	Meaning
Interaction	A pattern in which a comparison across one factor's levels changes with the other factor's level. A difference of differences expresses one such comparison.
Simple effect	A comparison across one factor's levels at a specified level of the other factor. It describes a conditional comparison within the factorial design.
Three-way interaction (higher-order interaction)	A pattern in which a two-way interaction changes across the levels of a third factor. Each added factor multiplies the cells and adds interactions, so the highest-order interaction is read first and a few comparisons are declared in advance.

Chapter 5: Description and Basic Inference with R

Table 5
Key terms of Chapter 5.

Term and common alternatives	Meaning
Script and Quarto source	A saved `.R` script records commands so an analysis can be rerun; a `.qmd` source connects prose, mathematics, executable code, tables, figures, and references, and renders to PDF, HTML, or Word.
Simulation and seed (Monte Carlo simulation)	Generating data from a declared process to see what a statistic or procedure does across repetitions. A seed makes the random draws repeatable; it does not make the process true for an actual study.
Monte Carlo standard error	The numerical uncertainty of a simulated probability or mean that comes from using finitely many repetitions; more repetitions shrink it. It is distinct from a study's SE, which depends on the study's sample size.
Covariance matrix	A square table of variances on the diagonal and covariances off the diagonal for a stated set of variables and observations.
Correlation matrix	A square table of pairwise correlation coefficients for a stated set of variables and observations; its diagonal is 1 for variables with positive variance.

Chapter 6: Group Comparisons and Statistical Modelling with R

Table 6
Key terms of Chapter 6.

Term and common alternatives	Meaning
Model formula (R formula)	Notation such as `score ~ group` naming an outcome and the variables that explain it for a fitting function. It does not state whether observations are independent or paired; that comes from the design.
Categorical effect scales (difference in proportions; risk ratio; odds ratio; Cramér's V)	Four ways to state the size of an association in a table, each answering its own question: the difference in proportions gives an absolute gap, the risk ratio a relative gap, the odds ratio a ratio of odds that stays the same when rows and columns exchange roles, and Cramér's V a unit-free strength, from 0 to 1, for a table of any size.
Objective function	A numerical criterion evaluated at candidate parameter values for fixed data; fitting seeks a stated minimum or maximum.

Chapter 7: Simple Linear Regression

Table 7
Key terms of Chapter 7.

Term and common alternatives	Meaning
Simple linear regression (simple regression model; regression line)	A model in which an outcome's mean is a straight line in one predictor, $E(Y \mid x) = \beta_0 + \beta_1 x$, with errors around the line that average zero and share one SD σ; its three parameters are the intercept, the slope and σ. The least-squares line estimates the model's line: its descriptive use concerns the records fitted, and a model-based use needs a declared process.
Intercept	The fitted outcome at predictor value zero. It places the line vertically; when zero is outside the observed support, it is a mathematical anchor rather than a supported prediction. Centering the predictor moves it to a supported reference point.
Slope (regression coefficient)	The difference in fitted outcome per one-unit difference in the predictor, in outcome units per predictor unit. It compares units that differ in the predictor; it is not a change within one unit unless a design supports that reading.

Term and common alternatives	Meaning
Comparison and counterfactual interpretations (predictive and causal readings)	A comparison reading treats a coefficient as the predicted difference in outcome between units that differ by one unit in the predictor; it is the default for observational records. A counterfactual reading treats it as a causal contrast between outcomes under two predictor values. No unit is observed under both, so a credible comparison, such as a randomized one, can identify the average contrast across units even when individual contrasts differ, but not one unit's own.
Ordinary least squares (OLS)	The rule that chooses the intercept and slope minimizing the residual sum of squares. With an intercept, the fitted line passes through the mean point and the residuals sum to zero; the slope equals covariance divided by predictor variance.
Residual sum of squares (RSS)	The sum of squared residuals from a fitted line, in squared outcome units. Total squared variation about the mean splits into a regression part and RSS when the model has an intercept.
Residual standard deviation (residual standard error; standard error of the estimate)	The square root of RSS divided by its residual degrees of freedom, in outcome units; with an intercept and one predictor the degrees of freedom are $n - 2$. Under the process model it estimates the error SD.
R-squared (coefficient of determination)	One minus RSS divided by total squared variation: the proportion of centered squared outcome variation accounted for by the fit in these records. In simple regression with an intercept it equals the squared correlation; for any least-squares fit with an intercept it equals the squared correlation between the outcome and its fitted values.
Confidence interval for a mean response (mean-response interval)	An interval for the process mean outcome at a stated predictor value, using the SE of the fitted value. It concerns an average, narrows near the predictor mean, and does not cover individual outcomes.
Prediction interval (new-case interval)	An interval for one new outcome at a stated predictor value from the same process. Its variance adds the error variance to the fitted-mean variance, so it is wider than the mean-response interval and keeps a nonzero floor as the sample grows.
Residual plot (residuals against fitted values)	A plot of each residual against its fitted value, read for its centre, where a curve questions the straight-line mean, and its spread, where widening scatter questions one error SD and the prediction interval built on it.
Extrapolation (predictor support)	Using a fitted line at predictor values outside the observed range or where observations are sparse. Software returns a number there; the data do not support it.
Regression toward the mean (regression to the mean)	For a simple linear prediction in standard units, $\hat{z}_y = r\, z_x$; when $\lvert r \rvert < 1$, the predicted value is nearer its mean than the predictor is to its mean. In a stable repeated-measurement model with independent occasion errors, a group selected on an extreme noisy first score is expected to be less extreme on the second. These are statements about predictions or conditional averages, not every case; a narrower spread of predictions does not imply a narrower outcome distribution (the regression paradox).

Term and common alternatives	Meaning
Regression fallacy	Attributing to an intervention the movement expected from selection on an extreme noisy measurement. Evaluating an intervention requires a credible no-intervention benchmark, supplied by a suitable design or untreated comparison under stated assumptions; a fitted line through treated outcomes alone does not supply it.

Chapter 8: Multiple Regression I: Adjusted Relationships, Group Comparisons, and Model Comparison

Table 8
Key terms of Chapter 8.

Term and common alternatives	Meaning
Multiple regression (multiple regression model; additive model)	A model in which an outcome's mean is an intercept plus one term per predictor, $E(y \mid x_1, \ldots, x_p) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$, with errors of one common SD σ; its $p+1$ coefficients leave $n-p-1$ residual df, and least squares estimates them. Each coefficient compares means at the same values of the other predictors; the additive form assumes each comparison is the same across those values.
Partial slope (conditional coefficient; adjusted coefficient)	The difference in fitted outcome per one-unit difference in one predictor, holding the model's other predictors fixed. Its meaning depends on the conditioning set named with it; it is not an intervention effect or a claim about unmeasured differences.
Confounding and statistical control (adjustment)	In the regression sense, confounding is a distortion of the relationship between a focal predictor and the outcome by a variable associated with both; statistical control includes such variables so that the focal coefficient compares units at the same values of them. Control yields an adjusted association for the measured, included variables in the form the model gives them, not automatically a causal effect.
Partial regression (residualization; added-variable form)	Regressing both the outcome and one predictor on the other predictors, then relating their residuals. The residual slope equals the partial slope; the residual correlation is the partial correlation, unit-free.
Coefficient covariance matrix	The estimated variances and covariances of the fitted coefficients (`vcov()` in R). Any linear combination of coefficients, such as a fitted mean or a contrast, has variance $\mathbf{c}^{\mathsf{T}}\widehat{\mathbf{V}}\mathbf{c}$; adding SEs or ignoring covariances is a different, incorrect calculation.
Adjusted R^2	R^2 with a penalty for the number of predictors, $1-(1-R^2)(n-1)/(n-p-1)$. It rises only when an added predictor's partial F exceeds 1; like R^2, it is an in-sample summary.
Overall F test (omnibus test)	Tests whether all slopes are zero, $F = \{RegSS/p\}/\{RSS/(n-p-1)\}$ on p and $n-p-1$ df. A significant result says the predictors together improve on one common mean, not that each predictor matters.

Term and common alternatives	Meaning
Nested-model F test (partial F; extra sum of squares; block test)	Compares a reduced model with a full one that contains it, fitted to the same records and outcome: the RSS reduction per added coefficient, divided by the full model's residual mean square, on q and $n - p_F - 1$ df. It tests the added block jointly, not each coefficient separately.
Likelihood-ratio test (deviance difference)	Compares nested models by twice the rise in maximized log-likelihood, $LR = 2(\hat{\ell}_F - \hat{\ell}_R)$, which for Normal linear models is $n \log(RSS_R/RSS_F)$, referred to a chi-square distribution with q df in large samples. For linear models the exact F test is preferred; the likelihood-ratio test extends to models without a residual sum of squares.
Information criteria (AIC; BIC)	Penalized in-sample fit: $-2\hat{\ell}$ plus $2K$ (AIC) or $K \log n$ (BIC), with $K = p + 2$ counting the coefficients and the error variance. Smaller is preferred among models fitted to the same records and outcome scale, nested or not; BIC charges more per parameter once $n \geq 8$, and neither is a held-out error.
Indicator variable (dummy variable) and reference category	A 0/1 variable marking membership in one category of a factor; a factor with J levels needs $J - 1$ indicators, and the omitted level is the reference. The intercept refers to the reference category; each indicator coefficient is a difference from it at the same other predictors. Releveling changes coordinates, not fitted values.
Joint support (covariate hull)	Whether a combination of predictor values lies within the region where observations actually occur, not merely within each variable's separate range. A fitted value outside joint support is extrapolation even if each coordinate is observed somewhere.
ANCOVA (analysis of covariance)	A regression comparing groups while including a quantitative covariate with one common slope; its adjusted group means are read at a stated covariate value within common support. The posttest-only, gain-score and ANCOVA analyses are one regression whose pretest coefficient is 0, 1 or estimated; adjustment cannot supply what random assignment supplies.
Homogeneity of regression slopes (parallel slopes)	ANCOVA's condition that the outcome–covariate slope is the same in every group, checked by testing the group-by-covariate product block; when the slopes differ, the adjusted difference depends on the covariate value at which it is read.
Standardization over a named distribution (model-averaged fitted value)	Averaging the model's fitted values after setting one predictor to a chosen value for every profile in a declared set with fixed weights. Its uncertainty uses the coefficient covariance; it is not an observed group mean, a causal effect, or held-out performance.

Chapter 9: Multiple Regression II: Model Assumptions, Diagnostics, and Respecification

Table 9
Key terms of Chapter 9.

Term and common alternatives	Meaning
Model checking (regression diagnostics)	Examining whether a fitted model's mean function and error assumptions are adequate for the answers drawn from it, mainly through residual plots and the design. It judges whether departures are minor, whether they change conclusions, and whether a better model is available; it is not a pass–fail ritual.
Regression assumptions (L.I.N.E.)	Linearity of the conditional mean, independence of the errors, Normal errors, and equal error variance at every combination of predictor values. They concern the unobserved errors, not the raw outcome or predictors, and are checked through residual plots and the design.
Standardized residual (Normal Q–Q plot; scale–location plot)	A residual divided by its estimated SD, $e_i/(s_e\sqrt{1-h_i})$, roughly standard Normal under the model. A Q–Q plot compares the ordered standardized residuals with Normal quantiles; a scale–location plot shows whether their spread changes with the fitted value. A pooled Q–Q plot cannot separate non-Normal errors from spread that differs across cases.
Leverage and influence (Cook's distance)	Leverage measures how unusual a case's combination of predictor values is. The leverages average $(p+1)/n$. Cook's distance measures the overall change in fitted values when a case is omitted. Its influence on a particular slope also depends on where the case lies on that predictor after adjustment for the others. Check influential cases and report how the fit changes when they are omitted.
Heteroscedasticity-consistent covariance (HC3; robust standard error)	A coefficient covariance that allows a different error variance for each case, using each squared residual adjusted for leverage. It can raise or lower a standard error and leaves the coefficients unchanged; it does not repair a wrong mean, dependence, or measurement error.
Multicollinearity (redundancy among predictors)	Strong linear association among predictors, so that one can be predicted well from the others. It leaves the joint fit intact but makes individual coefficients unstable, inflates their standard errors, and makes holding the other predictors fixed a comparison the records rarely show.
Variance inflation factor (VIF; tolerance)	$\text{VIF}_j = 1/(1-R_j^2)$, where R_j^2 comes from regressing x_j on the other predictors and $1-R_j^2$ is the tolerance. Holding error variance and predictor spread fixed, the VIF multiplies the slope variance relative to uncorrelated predictors. Cutoffs such as 5 or 10 are conventions.
Standardized coefficient (beta weight)	A partial slope expressed in outcome SDs per predictor SD, $\beta_j^* = \widehat{\beta}_j\, s_{x_j}/s_y$, with the other predictors held fixed. It compares predictors within one model and sample; it is not a causal ranking.
Variance partition (unique and shared contributions; change in R^2)	Dividing a model's R^2 into parts explained uniquely by each predictor, the change in R^2 when it is added last, and a part shared among predictors. A predictor's total part is its squared correlation with the outcome; unique parts can be small for redundant predictors.

Term and common alternatives	Meaning
Outcome transformation (log; square root; Box–Cox)	Refitting the model to a function of the outcome to stabilize its spread or reduce skew. The Box–Cox family $y^{(\lambda)}$ includes the log ($\lambda = 0$) and square root ($\lambda = 1/2$) and chooses λ by likelihood. A changed scale is a changed question, and fit statistics do not carry across it.
Log-transformed outcome (geometric-mean scale)	Fitting the logarithm of the outcome, so that a coefficient is an additive difference in expected log outcome and $100(e^{\beta} - 1)$ percent is a multiplicative difference in geometric means.
Polynomial term (quadratic term; curvature)	A power of a predictor added as a further column so that the fitted mean can curve while the model stays linear in its coefficients. The predictor's slope then depends on its value; centering before squaring moves the linear coefficient's reading to the centre without changing the fit.
Bootstrap (case resampling; percentile interval)	Refitting the model to many resamples of the records drawn with replacement and using the spread of the refitted estimates as an SE, or their middle 95% as a percentile interval. It treats the resampled units as independent and exchangeable; it does not repair dependence, thin support, measurement error, or an observational design.
Cluster-robust standard error (school-clustered covariance)	A coefficient covariance that sums residual contributions within clusters, such as schools, before combining them, allowing dependence within clusters under an approximation that treats clusters as independent. It changes uncertainty, not the coefficient, and does not repair missing outcomes or weighting.

Chapter 10: Interaction and Moderation

Table 10
Key terms of Chapter 10.

Term and common alternatives	Meaning
Interaction (product term; nonadditivity)	A model term formed by multiplying predictors, so that the slope of one depends on the value of another: in $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$, β_3 is the change in the slope of x_1 per unit of x_2 and equally the change in the slope of x_2 per unit of x_1. Nonadditivity is defined on the modeled scale; a nonlinear change of outcome scale can create or remove it.
Moderator (moderating variable; moderation)	The variable across whose values or groups a named focal relationship is allowed to vary. The product-term algebra is symmetric, so the label comes from theory, timing or design; when roles cannot be justified, symmetric language and both sets of simple slopes are used. A moderation reading describes when a relationship differs, not how an effect is transmitted.

Term and common alternatives	Meaning
Simple slope (conditional slope; conditional effect)	The slope of a focal predictor at a stated value of the moderator, such as $\beta_1 + \beta_3 w$. Its SE is the square root of $\mathrm{Var}(\hat{\beta}_1) + w^2\,\mathrm{Var}(\hat{\beta}_3) + 2w\,\mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_3)$, which combines the variances and the covariance of the coefficients involved; adding SEs or dropping the covariance is a different, incorrect calculation.
Pick-a-point approach (simple slope analysis; probing)	Estimating and graphing simple slopes at a few moderator values, such as the mean and one SD either side, the quartiles, meaningful cut points, or the two values of a 0/1 moderator. It reports magnitudes at interpretable values, but its significance statements depend on the probes chosen.
Johnson–Neyman technique (region of significance)	Finding the moderator values at which the pointwise interval for a simple slope touches zero, the roots of a quadratic. Within the observed range there can be one, two or no boundaries; a boundary outside the data has no empirical support, and a pointwise band is not a simultaneous band.
Hierarchy principle (marginality)	The rule that a model with a product term keeps every lower-order term the product contains, even when those coefficients are small or nonsignificant. Omitting one forces a slope to zero at an arbitrary origin and changes the fitted surface when a predictor is shifted.
Centering a predictor (mean-centering)	Subtracting a chosen value, often the mean, from a predictor so that the lower-order coefficients become slopes at that value. It changes the intercept and lower-order coefficients, and their SEs, while leaving the fitted values, the product's estimate and its test unchanged; it adds no information and does not reduce collinearity in a way that matters.
Difference of differences (interaction contrast)	A comparison of two simple effects or slopes, such as a status difference at one level minus that at another. In a regression with indicators and their products, a product coefficient is a difference of differences of cell means; its SE comes from the coefficient covariance.
Partial eta squared (η^2_{partial})	An effect's sum of squares divided by the sum of that effect's and the error sum of squares, $SS_{\text{effect}}/(SS_{\text{effect}} + SS_{\text{error}})$. It equals η^2 in a one-way design. With unequal cell sizes, a sequential main effect's sum of squares and its partial eta squared can depend on factor order.
Separate-slopes model (analysis of covariance with interaction)	A regression of an outcome on a factor, a covariate and their products, giving each group its own covariate slope. The products test whether the slopes differ; group differences are then read at a stated covariate value inside every group's observed range.
Higher-order interaction (three-way interaction; moderated moderation)	A product of three or more predictors, kept with all its lower-order terms. A three-way term asks whether the moderation of one predictor's slope by a second predictor changes across a third; it needs support in every cell it compares.

Chapter 11: Binary Logistic Regression

Table 11
Key terms of Chapter 11.

Term and common alternatives	Meaning
Generalized linear model (GLM)	A random component (the outcome's distribution), a linear predictor $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ and a link $g(\mu) = \eta$ connecting the conditional mean to it, with variance $\phi V(\mu)$; ordinary regression is the Normal, identity-link case, and changing the distribution and link gives the models of Chapters 11 and 12.
Event and non-event (the category coded 1)	The category a binary model treats as occurring, named with the category coded 0. Reversing the event reciprocates the odds ratios and complements the fitted probabilities; probabilities for the original event agree after relabeling.
Prevalence (event rate)	The proportion of analytic rows that are events. Precision also depends on conditional probabilities, predictor spread and overlap; an odds ratio approximates a risk ratio only when the compared risks are small.
Link function; logit and inverse logit (logistic function)	The logit is the log of the odds; its inverse is the S-shaped function bounded in $(0, 1)$, equal to one half and steepest where the linear predictor is zero. The link transforms the mean, not the observed 0 or 1.
Odds and log-odds	The odds are $\pi/(1-\pi)$ and the log-odds their logarithm; the three scales describe one quantity, with null values 1 and 0, and the log-odds scale is symmetric under reversing the event.
Odds ratio (OR)	The factor e^{β_j} by which the odds differ between cases one unit apart in a predictor and agreeing on the others; it depends on the units, reference, event and adjustment set, is not a risk ratio unless the event is rare, and implies no constant probability difference.
Divide-by-4 rule	For a constant log-odds slope β_j, the probability slope $\beta_j \pi(1-\pi)$ is steepest at $\pi = .5$. Its value there is $\beta_j/4$, and its magnitude bounds the absolute probability difference for a one-unit step.
Noncollapsibility (of the odds ratio)	A conditional odds ratio generally differs from the marginal one even without confounding, so a coefficient that moves under adjustment is not by itself evidence of confounding.
Likelihood and maximum likelihood (log-likelihood; MLE)	The probability of the observed data under a model, viewed as a function of its parameters with the data fixed, and the choice of parameter values that maximize it or its logarithm; it is not a probability distribution over the parameters, and the curvature of the log-likelihood at its maximum gives the large-sample SE.
Bernoulli likelihood and score equations	For independent binary outcomes, the likelihood is the product of π_i over events and $1-\pi_i$ over non-events. The logistic score equations $\sum_i x_{ij}(y_i - \pi_i) = 0$ generally require numerical solution. With an intercept, the fitted probabilities sum to the observed event total.
Iteratively reweighted least squares (IRLS)	The algorithm that repeatedly solves a weighted linear regression with weights $\hat{\pi}_i(1-\hat{\pi}_i)$ until the coefficients stop changing; convergence establishes no adequacy.

Term and common alternatives	Meaning
Deviance and the likelihood-ratio test (G^2)	For ungrouped binary data the deviance is -2 times the maximized log-likelihood and has no fit reading alone; the difference between nested models on the same rows and event, $G^2 = 2(\hat{\ell}_{\text{full}} - \hat{\ell}_{\text{reduced}})$, is referred to chi-square.
Wald test and profile-likelihood interval	The Wald statistic divides an estimate by its SE, and its interval is exponentiated for an odds ratio; a profile interval collects the values a likelihood-ratio comparison does not reject. They diverge for small samples, rare events or near-infinite estimates.
McFadden pseudo R^2	One minus the ratio of the fitted to the intercept-only maximized log-likelihood: not a proportion of variance explained, and routinely small for useful models.
Ratio of odds ratios (interaction on the odds scale)	The exponential of a product coefficient: the factor by which one predictor's odds ratio changes per unit of the other. It is read through conditional odds ratios at stated values and predicted-probability plots, with the scale named.
Empirical logit	The log-odds of a grouped event count with one half added to each count, plotted against a predictor to check that the log-odds is linear in it.
Binned residual	The mean raw residual within a group of cases with similar fitted probabilities, against an approximate two-SE band; it replaces the residual-versus-fitted plot, which shows only two bands for binary data.
Separation (complete; quasi-complete)	A predictor pattern separates events from non-events completely or on a boundary, so at least one coefficient has no finite maximum-likelihood estimate; diverging coefficients, huge SEs and many iterations are symptoms, and no single warning is required.
Calibration and discrimination	Calibration asks whether fitted probabilities match observed frequencies, discrimination whether they rank events above non-events; they answer distinct questions, and new-case performance needs held-out data.
Linear probability model (LPM)	Least squares on a 0/1 outcome. An additive slope gives a fitted probability difference per unit. Fitted values can leave $[0, 1]$, and the conditional variance $\pi(1 - \pi)$ generally varies across predictor values.
Averaged probability difference (average marginal comparison)	The mean, over the analytic rows with their own other predictor values, of the difference between fitted probabilities at two values of one predictor.

Chapter 12: Extending the GLM: Nominal, Ordinal, and Count Outcomes

Table 12
Key terms of Chapter 12.

Term and common alternatives	Meaning
Outcome triage	The measurement judgment, made before any model is fitted, classifying an outcome as binary, nominal, ordered, a count, a rate, a count with many zeros or a proportion of known trials. The model follows the classification and cannot establish it.
Multinomial logistic regression (baseline-category logit; generalized logit)	For J unordered categories, $J-1$ log-ratio equations against one baseline category, each with its own intercept and slopes, estimated jointly so that the category probabilities sum to one. The fitted probabilities are the same whichever baseline is chosen; the coefficients are not.
Baseline outcome category	The outcome category in the denominator of every multinomial equation: a coordinate choice, distinct from the reference level of a categorical predictor, whose change alters every reported ratio.
Relative risk ratio (RRR)	The exponentiated multinomial coefficient: the factor by which the ratio of one category's probability to the baseline's is multiplied per predictor unit. It is not a probability difference, and is incomplete unless both compared categories are named.
Independence of irrelevant alternatives (IIA)	The multinomial model's assumption that the probability ratio of two categories does not depend on which other categories are available: a claim about the choice set, argued from the design.
Cumulative probability and cumulative logit	$P(y \le k)$ at each of the $J-1$ cut points, and its log-odds. A category probability is a difference of two adjacent cumulative probabilities.
Threshold (cut point; ordinal intercept)	One of the ordered constants $\zeta_1 < \cdots < \zeta_{J-1}$ placing the cut points on the logit scale; they are location constants, not effects.
Proportional odds (common-slope restriction; partial proportional odds)	The restriction that one slope vector governs every cut point, so one odds ratio applies above every cut. It is imposed by the analyst, not conferred by ordering the levels, and is testable against cut-specific slopes; a partial model frees selected slopes.
Poisson regression (log-linear count model)	A GLM with a Poisson outcome and a log link, so the expected count is positive and differences are multiplicative; its conditional variance is forced to equal the mean.
Incidence-rate ratio (IRR; rate ratio)	The exponentiated coefficient of a log-link count or rate model: the factor by which the expected count, or rate with an offset, differs per predictor unit. An absolute difference also needs its baseline.
Exposure and offset	The window over which each unit was observed, and $\log t_i$ entered with its coefficient fixed at one so the model describes a rate. An offset is a decision about the estimand, with no SE or p-value; freeing its coefficient gives a testable alternative.

Term and common alternatives	Meaning
Overdispersion and the dispersion statistic	Conditional variance exceeding the conditional mean, assessed by the mean squared Pearson residual $\hat{\phi}$ under a suitable mean model. Under a constant variance-to-mean multiplier greater than one, the usual Poisson SEs are too small. The marginal variance also reflects differences in the conditional means.
Quasi-Poisson (quasi-likelihood)	The Poisson mean with variance $\phi\mu$: identical estimates and every SE multiplied by $\sqrt{\hat{\phi}}$. No full distribution is specified, so there is no log-likelihood, AIC or BIC.
Negative binomial regression	A count model with variance $\mu + \mu^2/\theta$, arising when the Poisson mean varies across units sharing predictor values. It is a different likelihood, not a rescaling, and the Poisson model is its boundary case $1/\theta = 0$.
Zero-inflated model	A mixture of an always-zero component, modeled by a logit, and a count component that can also produce zeros. The count-component mean differs from the overall mean, and an observed zero does not reveal which component produced it. The variance-to-mean ratio, $1 + \pi\lambda + \lambda/\theta$ (ZIP: without the last term), shows that excess zeros alone create overdispersion.
Hurdle model (two-part count model)	A binary model for zero versus a positive count combined with a zero-truncated count distribution for the positive values; the overall mean combines the probability of a positive count with the positive-count mean.
Zero-truncated count distribution	A count distribution conditioned on a positive outcome: its zero probability is removed and the positive probabilities rescaled. Its mean is the original mean divided by the probability of a positive count.
Loglinear model	A Poisson model for the cell counts of a cross-classified table, with the classifying variables as predictors. Its two-factor terms are conditional log odds ratios, and with one variable as the response it reproduces the corresponding multinomial or logit model.
Likelihood-ratio statistic G^2 (deviance of a table)	$2\sum O\log(O/E)$ over the cells of a table, the deviance of a loglinear model, referred to chi-square on its residual df when the expected counts are adequate. The Pearson X^2 uses the same expected counts.
Conditional independence	Two variables unrelated within every level of a third, as in the model [SF][SY]; their association in the table with the layers added can still differ from zero.
Homogeneous association	The same odds ratio between two variables in every layer of a third: the loglinear model with all two-factor terms and no three-factor term.

Chapter 13: Prediction, Cross-Validation, and Regularization

Table 13
Key terms of Chapter 13.

Term and common alternatives	Meaning
Generalization error (prediction error; risk)	Expected loss on a new unit from the stated deployment distribution. It can be defined for one fitted model conditional on its training data, or averaged over training samples for a learning rule at a stated training size; held-out observations estimate the chosen target.
Deployment moment (prediction time)	The point at which a prediction must be produced, which decides whether a predictor is available. A variable recorded later may improve measured performance but cannot be used for that earlier prediction task.
Apparent error and optimism (training error; resubstitution error)	Loss on the records used to fit the rule. Using those outcomes in fitting generally makes this loss optimistic for new-case performance; the gap to held-out loss is the optimism, which greater flexibility can increase, without a guaranteed ordering in every sample or for every loss.
Root mean squared error (RMSE)	The square root of the mean squared prediction error, in outcome units. It penalizes a large error far more than MAE does, so the choice between them states which mistakes matter.
Out-of-sample R^2	One minus the held-out squared error divided by the held-out squared error of a named constant reference fitted to the same training rows. It can be negative and is not the in-sample R^2 of Chapter 7.
k-fold cross-validation	Splitting the development records into k folds, fitting on $k-1$ of them and predicting the withheld fold in turn, so that every record is predicted once by a fit that did not use it. It estimates the risk of a learning rule at training size about $(k-1)n/k$; the folds' training sets overlap.
Data leakage (information leakage)	Using information unavailable at the deployment moment, or letting held-out records influence a training or selection step used to predict them. It can make performance estimates optimistic; there is no general proportional rule for the size of that optimism.
Development and test split (holdout set)	A partition in which fitting, screening and tuning use the development records, while test outcomes remain protected until the rule is fixed. Adapting the rule after seeing test results uses the test set for development; repeating an unchanged calculation does not itself do so.
Penalized regression (shrinkage; ridge; lasso; elastic net)	Fitting that adds a coefficient penalty to the loss, trading bias against variance. Ridge's squared penalty generally shrinks coefficients without selecting exact zeros; the lasso's absolute-value penalty can set coefficients to zero, and elastic net combines the penalties.
Bias–variance trade-off	The split of expected squared prediction error into noise variance, squared bias and variance across training samples. A penalty can add bias and remove more variance, lowering the total.
Tuning parameter (penalty; λ)	A quantity that indexes a family of fits and is chosen from data the fit did not see. Choosing it is itself a data-dependent step and belongs inside the resample.

Term and common alternatives	Meaning
One-standard-error rule	A convention choosing the largest penalty whose observed cross-validated loss is within the estimated standard error at the minimum. It prefers a simpler fit under that empirical tolerance; it does not bound the increase in true prediction error.
Nested cross-validation	Cross-validation with an outer loop that assesses a whole selection or tuning procedure and an inner loop, inside each outer training part, that makes the selection. It estimates how the procedure, not one chosen rule, performs.
Proper scoring rule (log loss; Brier score)	A score for a predicted probability that is minimized in expectation by the true probability, so that honest uncertainty is rewarded. Accuracy and a single confusion matrix are not proper scores.
Calibration	Agreement between predicted probabilities and observed event rates among cases with similar predictions. It is invisible in a ranking measure such as AUC and is checked on held-out predictions.
Discrimination (AUC; concordance)	The probability that a randomly chosen event case receives a higher score than a non-event case, with half credit for tied scores. It depends on ranking and does not establish calibration of the probabilities.
Decision threshold (operating point)	The predicted probability above which an action is taken, determined by the ratio of the costs of the two errors and not by the model. Two decision-makers with different costs read the same probabilities and act differently.
Confusion matrix (sensitivity; specificity; precision; accuracy)	The two-by-two table crossing a rule's actions with the outcomes. Sensitivity is the share of events flagged, specificity the share of non-events left unflagged, precision the share of flags that are events, and accuracy the share of correct decisions, which depends heavily on how common the event is.
Dataset shift	A difference between the distributions in the development and use settings, potentially involving predictors, outcome frequencies or their relationship. Internal resampling alone does not establish performance in a different setting.

Chapter 14: Regression Pitfalls and Causal Design

Table 14
Key terms of Chapter 14.

Term and common alternatives	Meaning
Directed acyclic graph (causal diagram; DAG)	A diagram whose arrows state assumed direct causes and whose directed paths cannot return to their starting point; it records assumptions that can be questioned, rather than conclusions produced by the data.
Mediator (intermediate variable)	A variable caused by the focal predictor and lying on a path to the outcome; one kind of post-treatment variable, not a synonym for every variable measured after treatment. Conditioning on it changes a total-effect comparison; a direct causal reading needs additional identification and modeling conditions.
Collider (common effect)	A variable caused by two others. A path through a collider is blocked unless the collider or a descendant is conditioned on; such conditioning can induce an association along that path.
Adjustment set	The variables conditioned on for a stated estimand under an assumed structure, justified by that structure and the timing rather than by the available columns or a coefficient's movement.
Aggregation reversal (Simpson's paradox; strict Simpson reversal)	A sign difference between a pooled comparison and the stratified or standardized comparisons, strict when every stratum points the opposite way to the pooled result; it arises from composition, not from a computational error.
Within-group, between-group, and pooled comparison	Three estimands from one dataset: inside a level of the grouping variable, across group summaries, and across all rows ignoring the grouping, with the pooled slope a weighted average of the other two by the within and between shares of predictor variation.
Direct standardization (standardized comparison)	Recomputing each group's summary under one declared common distribution of a stratifying variable; a third estimand rather than a compromise between the pooled and conditional comparisons.
Range restriction (restriction of range)	Keeping only part of the observed range; restricting the predictor under a correctly specified linear model costs precision, apparent strength and support outside the window, while restricting on the outcome also moves the slope.
Missingness mechanisms (MCAR; MAR; MNAR)	MCAR means missingness is unrelated to observed and missing values; MAR means it is unrelated to missing values conditional on observed information; MNAR means dependence on missing values remains. Complete-case validity depends on the target and mechanism, and MNAR requires assumptions not identified by the observed data alone.
Attenuation from measurement error	In a single-predictor linear model, classical predictor error multiplies the population slope by reliability. With several predictors the direction can differ; a mismeasured control can leave residual confounding.

Term and common alternatives	Meaning
Ecological inference (ecological fallacy)	Inferring an individual-level relationship from aggregate records. Aggregate and individual relationships can differ in sign and magnitude; the aggregate relationship alone does not identify the individual comparison.
Specification sensitivity (multiverse analysis; analyst degrees of freedom)	The spread of an estimate across the defensible analyses that could have been run; a disclosure about specification, not evidence about an omitted common cause, selection or measurement shared by every version.
Pseudoreplication (dependence as a design fact)	Treating dependent observations as independent replicates, misrepresenting uncertainty. The relevant dependence, sampling and assignment structure must be represented; its effect on a standard error depends on the comparison.
Identification argument (design warrant)	The explicit reasons a comparison estimates the intended causal quantity, whether assignment or a named quasi-experimental design with its stated assumption; naming a design is not meeting its assumption.
Bounded regression claim (credible-claim template)	A concluding statement whose unit, verb, conditioning set, supported range, design and disclosed sensitivity all match the analysis that produced it.

Argument Chapter A: How to Read a Quantitative Study as an Argument

Table 15
Key terms of Argument Chapter A.

Term and common alternatives	Meaning
Argument component	One of the four functions, evidence, justification, inference, and bounded claim, that a piece of study information performs in the statistical or the substantive layer. A reader identifies the function before judging its adequacy.
Documentary location	The place in a report or its related materials, such as Methods, a protocol, a codebook, or an analytical supplement, where the traceable information for a component is found.
Evaluation question	One of the six questions on alignment, support, and translation used to judge the quality of a question-level argument after it has been read. Each receives a reasoned judgment, not a score.

Argument Chapter B: Extending the Argument-Based Approach to Quantitative Studies

Table 16
Key terms of Argument Chapter B.

Term and common alternatives	Meaning
Statistical strand	An estimand-specific statistical chain inside one argument. Two strands arise only when sibling estimands are jointly required for one integrated substantive answer; each has its own statistical inference and conclusion.
Coordinated questions	Several question-answer pairs in one study whose support depends on one another. Each answer is read on its own, and their coherence across questions is judged separately.

Procedures and Their Conditions

The table below gathers, for each procedure that Chapters 3 and 4 teach, what its result estimates and the conditions it relies on. *Secured by design* lists the conditions a study's design can supply, such as random assignment, random sampling, or pairing; *Model assumptions and how to check them* lists the conditions the statistical model adds and the evidence available about each; *If a condition fails* gives the response the chapters teach; and *Chapter* names the chapter that develops the procedure. The choices made for one particular analysis, such as the sidedness of a test, the confidence level, or the family of comparisons, stay with that analysis in its chapter.

Table 17
Procedures and their conditions.

Procedure and target	Secured by design	Model assumptions and how to check them	If a condition fails	Chapter
One-sample t test: the mean μ of the sampled population, in outcome units, compared with a stated null value	Random sampling from the named population; independent units, each measured once	A Normal population gives an exact t reference; otherwise the sample mean needs a near-Normal sampling distribution. Look for strong skew, heavy tails or one dominating value in the scores; no sample-size cutoff settles the question	Dependent units: an SE that represents the pairs or clusters. Strong skew with few units: disclose the weaker approximation and compare a bootstrap interval. Without random sampling, a population reading needs a stated model	Ch. 3

Procedure and target	Secured by design	Model assumptions and how to check them	If a condition fails	Chapter
Paired-samples t test: the mean within-pair difference μ_D, in outcome units, with the subtraction direction stated	Pairs formed by the design, such as one student measured twice or two classes in one school; independent pairs. Random assignment within pairs supports a causal reading	Near-Normal pair differences give the t reference: inspect the differences themselves rather than the two sets of scores; a few pairs say little about shape	Ignoring the pairing: an independent-groups analysis omits the within-pair covariance and misstates the SE. No untreated comparison: a mean change alone cannot be separated from time trends or repeated testing. Skewed differences: resample whole pairs	Ch. 3
Equal-variance two-sample t test: a difference between two means, in outcome units, either a mean assignment effect over the participants or a difference between two populations' means	Random assignment of individuals, supporting a causal comparison among the participants, or two independent random samples; different units in the two groups	One common population variance and a near-Normal sampling distribution of the difference: compare the group SDs with the group sizes and look for skew or dominating values in each group. After random assignment the t reference is approximate. No interference between units is argued from the setting; the data cannot check it	Unequal spreads with unequal group sizes: the pooled SE is too small when the smaller group varies more, so choose Welch's t in advance from the design, not from a preliminary variance test. With equal group sizes the two SEs coincide	Ch. 3
Welch's unequal-variance t test: a difference between two means, in outcome units, when the two groups' variances may differ	Two independent random samples or randomly assigned groups, with different units in each group	No common variance is assumed. Each group mean needs a near-Normal sampling distribution, and the Welch–Satterthwaite df approximate the reference: inspect each group's distribution, especially the smaller group's	Skew or a dominating value in a small group: disclose the weaker approximation and compare a bootstrap that resamples within each group. Paired or clustered units: an SE that represents them	Ch. 3

Procedure and target	Secured by design	Model assumptions and how to check them	If a condition fails	Chapter
Pearson chi-square test of association: whether two categorical variables are associated in the stipulated population or process, against the null of no association (independence); conditional proportions give the direction and size	Independent sampling of units, each counted once in one of mutually exclusive categories	Adequate expected counts for the chi-square approximation: calculate every expected count, row total times column total divided by n, before reading the p-value	Small expected counts: an exact conditional test such as Fisher's (Ch. 6). Repeated or clustered records: the counts overstate the independent information. An observed rather than assigned grouping supports association only	Ch. 3
One-way ANOVA F test: whether J group means are all equal, an omnibus question about the whole pattern of means	Random assignment of individuals to the groups, or independent random samples; each unit in one group, measured once	Independent Normal errors with a common variance around the group means: compare the group SDs with the group sizes, and inspect residuals against fitted means and in a Normal quantile plot. Normality concerns the residuals, not the pooled scores	Unequal variances, above all with unequal group sizes: Welch's one-way ANOVA, chosen in advance. Classes or repeated measurements: an analysis that represents that structure. Missing outcomes: compare assigned and analysed counts and state what the analysis assumes	Ch. 4
Contrasts with a declared family: comparisons $C = \sum_j c_j \mu_j$ with weights summing to zero, in outcome units, with simultaneous coverage for the declared family	Random assignment of individuals to the groups	The one-way model's common variance and Normal errors, because every SE uses MS_W on $N - J$ df: check the group spreads and residuals as for the F test	Unequal spreads: MS_W misstates the SE of every contrast, tending to understate it for contrasts that include the more variable groups and to overstate it for those that exclude them; take each contrast's SE from the groups' own variances, which for a pair is Welch's t (Ch. 3). Comparisons chosen after inspection: a family that covers the selection, such as Scheffé's for all contrasts	Ch. 4

Procedure and target	Secured by design	Model assumptions and how to check them	If a condition fails	Chapter
Two-way (factorial) ANOVA: main effects, interaction and simple effects of two crossed factors, in outcome units, with the averaging weights stated	Random assignment of individuals to the cells of a crossed layout; one score per unit	Independent Normal errors with one common variance in every cell: inspect residuals against the fitted cell means; a few scores per cell say little about shape. A main effect is an average over the other factor's levels; reading it as the effect at every level assumes additivity: examine the interaction row and the cell-mean plot first	Interaction present: report cell means and simple effects rather than main effects alone. Unequal cell counts: state the averaging weights and the hypothesis each sum of squares tests. Unequal cell spreads, above all with unequal cell counts: treat the cells as the groups of one factor and take each stated contrast's SE from the cells' own variances; the joint F tests keep the common-variance model, a limit the report discloses	Ch. 4
Welch's one-way ANOVA: whether J group means are all equal when the group variances may differ	Independent groups of different units, randomly assigned or randomly sampled	No common variance is assumed: each group mean, weighted by its own precision, needs a near-Normal sampling distribution, and the approximate F reference estimates its denominator df from the group variances. Inspect each group's distribution and size	Clustered or repeated observations: an analysis that represents that structure. Very small or skewed groups: disclose the weaker approximation. Its scope is one factor: crossed factors with unequal cell spreads use contrasts among the cells with the cells' own variances	Ch. 4

Shared Notation

The table below lists the notation that two or more chapters share, with each symbol's meaning, the chapter where it is mainly used, and the alternatives or other symbols it must be kept apart from. Each chapter explains its symbols where they are first used, including the symbols that appear in one chapter only.

Table 18
Notation shared by two or more chapters.

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Data	n, n_1, n_0, n_j, N	Number of cases in the analysis sample; group sizes; a population or record count	Ch. 2	Named groups take named subscripts, such as n_A and n_B
Data	i, j, k; x_{ij}	Locally declared indices; a data-matrix entry	Ch. 2	In the data matrix, i is case and j variable. ANOVA declares factor/group/replicate indices beside its own model.
Data	y_i, x_i	Observed outcome and predictor values of case i; lowercase marks observed values	Ch. 2	y_{ij} for case i in group j
Data	Y_i, Y_{ij}	The outcome as a random variable before observation; uppercase marks random quantities	Ch. 3	Appear where the distinction from observed values matters; in Y_{ij} the second index is the group of case i, whereas the potential outcomes Y_{i1}, Y_{i0} have their own Estimands row
Summaries	$\bar{y}$, $\bar{x}$, $\bar{y}_j$, $\bar{d}$	Observed means; a group mean; the mean of paired differences	Ch. 2	Lowercase marks observed means, including group means
Summaries	$\bar{Y}$	The sample mean as an estimator in variance and SE derivations	Ch. 3	Reserved for the random quantity
Summaries	s, s^2, s_1, s_0, s_p, s_p^2, s_d	Sample SD and variance; group SDs; pooled SD and pooled variance; SD of paired differences	Ch. 2; Ch. 3	SD in prose; s_{xy} the sample covariance; the pooled SD and variance enter in Chapter 3, and Chapter 4's MS_W extends s_p^2 to k groups

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Summaries	r, ρ	Sample and population correlation	Ch. 2	A partial correlation carries its conditioning set in a subscript (row below)
Relationships	$r_{xy \cdot z}$	Sample partial correlation controlling linearly for z	Ch. 2	Residual association, not causal exclusion
Population	μ, σ, σ^2	Process or population mean, SD, and variance; $\mu_1 - \mu_0$ a mean difference	Ch. 2	$\mu_0 = \beta_0 + \beta_1 x_0$ the mean response at x_0 (Ch. 7), distinct from the group mean μ_0 in $\mu_1 - \mu_0$
Population	p, p_1, p_0	A population proportion and group proportions	Ch. 6	Distinct from the p-value, written p-value in prose and $p = .0059$ for its value
Estimands	Y_{i1}, Y_{i0}	Participant i's potential outcomes: the score participant i would have under assignment 1 and under assignment 0; over the n stated participants the mean assignment effect is $\Delta\mu = \frac{1}{n}\sum_{i=1}^{n}\{Y_{i1} - Y_{i0}\}$	Ch. 3; Ch. 5; Ch. 14	Each participant reveals only the outcome of the assigned condition; the second index names an assignment, not the observed group of Y_{ij}
Estimands	$\Delta\mu$, $\widehat{\Delta\mu}$	A difference in means as an estimand, $\Delta\mu = \mu_1 - \mu_0$ or the mean assignment effect over the stated participants, and its estimate $\widehat{\Delta\mu} = \bar{y}_1 - \bar{y}_0$; a stipulated value of $\Delta\mu$ serves planning and power calculations	Ch. 3; Ch. 14	$\Delta p = p_1 - p_0$ for a difference in proportions (Ch. 6); θ, $\hat{\theta}$ for a generic estimand and estimator, distinct from the negative-binomial θ of Chapter 12; τ and a bare Δ are not used
Uncertainty	$SE(\cdot)$, $\widehat{SE}(\cdot)$	The standard error of an estimator and its estimate; the hat marks the estimate. Name the statistic: $SE(\bar{y})$, $SE(\bar{y}_1 - \bar{y}_0)$, $SE(\widehat{C})$	Ch. 3	The hat marks an estimated SE; a chapter that abbreviates the notation says so where it does

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Uncertainty	$\mathrm{Var}(\cdot)$, $\mathrm{Cov}(\cdot,\cdot)$, $E(\cdot)$	Variance, covariance, and expectation; upright operator names with round brackets. Chapter 2 defines $E(X)$ for a probability model; Chapter 3 applies the same expectation to an estimator over stated repetitions, $E_{\text{assign}}(\cdot)$ when the repetition needs naming	Ch. 2; Ch. 3	$E[\cdot]$ and $\mathbb{E}$ are not used
Probability	$P(\cdot)$	Probability of an event	Ch. 2	—
Distributions	λ	A Poisson rate	Ch. 2; Ch. 12	Chapter 12 writes a Poisson model's expected count as λ and, with exposure t_i, a unit's rate per unit exposure as λ_i, so that its expected count is $t_i\lambda_i$; Chapter 6 uses λ for the noncentrality of a t distribution, Chapter 9 for the Box–Cox power, and Chapter 13 for a penalty size
Design	m, ICC, deff	Units per cluster, such as students per sampled school; the intraclass correlation, the share of outcome variance lying between clusters; the design effect $\text{deff} = 1 + (m-1)\,\text{ICC}$ for a mean from equal clusters	Ch. 2; Ch. 3	Chapter 2 introduces all three and Chapter 3 uses deff for a clustered SE. ICC and deff are upright abbreviations, not products of letters; Chapter 14 writes the intraclass correlation as ρ_I
Simulation	B	Number of simulation repetitions or bootstrap resamples, distinct from a study's sample size n	Ch. 5; Ch. 9	Monte Carlo SE shrinks with B; the study SE does not
Reference distributions	z, z^*	A standardized score and a standard Normal critical value	Ch. 2; Ch. 3	—
Reference distributions	t_{df}, t^*	The t distribution with df degrees of freedom; a critical value	Ch. 3	t_{n-2} in regression

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Reference distributions	χ^2, $F(df_1, df_2)$	Chi-square and F reference distributions with their degrees of freedom	Ch. 3; Ch. 4	—
Testing	X^2	Observed Pearson chi-square statistic	Ch. 3	Distinguish it from the χ^2_{df} reference distribution
Reference distributions	df	Degrees of freedom, upright; subscripts only when two are needed	Ch. 2; Ch. 3	Fractional Welch degrees of freedom follow the precision rule of the *Displayed values* row
Testing	H_0, H_A, α, β, $1-\beta$	Null and alternative hypotheses; Type I and Type II error probabilities; power at a specified alternative	Ch. 3	β here is distinct from a regression coefficient, and α from the elastic-net mixing parameter of Chapter 13
Magnitude	d, g	Standardized mean difference (Cohen's d, Hedges' g)	Ch. 3	—
ANOVA	SS_B, SS_W, SS_T, MS_B, MS_W; SS_A, SS_{AB}, SS_E, MS_E	Sums of squares and mean squares between, within, and total; factor, interaction and error terms of the two-way table; subscripted plain letters	Ch. 4	In the two-way table SS_B denotes factor B, not all between-cell variation
ANOVA	α_j, β_k, $(\alpha\beta)_{jk}$	Factor-model main effects and interaction departures in the two-way model	Ch. 4	Distinct from the unindexed Type I and Type II error probabilities α and β. Chapter 10 writes the second factor's effects as γ_k and the interaction as $(\alpha\gamma)_{jk}$, keeping β for regression coefficients
ANOVA	η^2	Eta squared, an ANOVA effect-size measure	Ch. 4	—
ANOVA	c_j, C, $\widehat{C}$	Contrast coefficients, a contrast $C = \sum_j c_j \mu_j$ with $\sum_j c_j = 0$, and its estimate $\widehat{C} = \sum_j c_j \bar{y}_j$	Ch. 4	L is not used for contrasts; $L(\cdot)$ is a likelihood. Chapter 8 writes a contrast of regression coefficients as $C = \mathbf{c}^\mathsf{T}\boldsymbol{\beta}$ with estimate $\widehat{C} = \mathbf{c}^\mathsf{T}\widehat{\boldsymbol{\beta}}$, and Chapter 10's Appendix A uses the same $\mathbf{c}^\mathsf{T}\boldsymbol{\beta}$ form

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
ANOVA	DD, $\widehat{DD}$	A difference of differences between group means and its estimate	Ch. 6	Chapter 4 develops the object as an interaction contrast declared in advance; Chapter 10 names the same object in words and reads it as a product coefficient of an indicator regression, with its SE from the coefficient covariance
Regression	β_0, β_1, β_j; $\hat{\beta}_0$, $\hat{\beta}_1$	Process intercept and slopes and their OLS estimates	Ch. 7	—
Regression	$\hat{y}_i$, e_i, ε_i	Fitted value, residual, and error term of case i	Ch. 4; Ch. 7	Chapter 4 uses the grouped forms $\hat{y}_{ij}$, e_{ij}, ε_{ij}
Regression	T_i; ε_{i1}, ε_{i2}	A case's stable true score and the independent errors of two measurements of it	Ch. 7; Ch. 14	Distinct from the residual e_i
Regression	S_{xx}, S_{xy}	Centered sums of squares and cross-products	Ch. 7; Ch. 14	—
Regression	RSS, $RegSS$, TSS	Residual, regression, and total sums of squares in regression; upright multi-letter names	Ch. 7	Distinct from the ANOVA SS family
Regression	s_e	Residual standard deviation (residual standard error)	Ch. 7	$\hat{\sigma}_{\mathrm{ML}}$ is the maximum-likelihood scale, a different estimate (Ch. 8)
Regression	R^2	Coefficient of determination	Ch. 7	Adjusted R^2 named in words

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Regression	D, D_i, D_j	Indicator (0/1) variables; D_i marks case i's membership, and $J-1$ indicator columns D_j code a J-level factor with a named reference level	Ch. 2; Ch. 6	Chapter 6 writes a factor's indicators with level subscripts, such as D_I and D_L. Chapters 8 and 10 write indicators as predictors, x with a level subscript such as x_{prof} or x_{A2}, with coefficients such as β_{prof}. Distinct from the deviance D (Chs. 11–12), Cook's distance D_i (Ch. 9), the group difference $D_j(x)$ (Ch. 10), and the diagonal SD matrix $\mathbf{D}$ (Ch. 5)
Regression	$\mathbf{X}$, $\mathbf{y}$, $\mathbf{H}$, $\widehat{\boldsymbol{\beta}}$	Design matrix, outcome vector, hat matrix, and coefficient vector; bold, with $\mathbf{X}^\mathsf{T}\mathbf{X}$ for the cross-product	Ch. 8; Ch. 9; Ch. 11	Matrices and vectors are bold, with a superscript T for a transpose; an element such as s_{jk} is an italic scalar. $h_{ii} = \mathbf{x}_i^\mathsf{T}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{x}_i$, with $\mathbf{x}_i$ case i's row of $\mathbf{X}$ written as a column, including an initial 1 for the intercept
Uncer-tainty	$\widehat{\mathbf{V}}$	Estimated covariance matrix of the fitted coefficients	Ch. 8	Distinct from the covariance matrix $\mathbf{S}$ of observed variables (Ch. 5)
Checks	h_i, h_{ii}, D_i	Leverage and Cook's distance of case i	Ch. 9	Chapter 7 defines leverage and Cook's distance in words for a one-predictor line; Chapter 9 gives formulas for several predictors, writing the leverage as h_i in the main text and as h_{ii}, the diagonal entry of $\mathbf{H}$, in Appendices A and B, and Cook's distance as D_i

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Likelihood	$L(\cdot)$, $\ell(\cdot)$, AIC, BIC	Likelihood, log-likelihood, and information criteria	Ch. 8; Ch. 9; Ch. 11	Chapter 8 supplies a Normal likelihood for its likelihood-ratio test and criteria and compares models with them, Chapter 9 reads them for its respecified models, Chapter 11 develops likelihood fully, and Chapter 12 carries it to its nominal, ordinal and count models. Chapters 8, 11 and 12 write the criteria as $-2\ell + 2K$ and $-2\ell + K \log n$ at the maximized log-likelihood, with K counting every estimated parameter
Design/contrasts	δ	A difference stipulated by the analyst: the population difference of one comparison in a power calculation (Ch. 6), or an assumed shortfall among missing outcomes in a sensitivity analysis (Arg. B)	Ch. 6; Arg. B	Each use is defined where it appears; a contrast of regression coefficients is written $C = \mathbf{c}^{\mathsf{T}}\boldsymbol{\beta}$ (ANOVA row c_j, C, $\widehat{C}$); a causal interpretation requires design and assumptions
Argument diagrams	E, J, I, C	Evidence, justification, inference and bounded claim: functions of information	Ch. 1; Arg. A	They name functions, not report headings or quality scores
Argument diagrams	STAT-E/J/I/C; SUB-E/J/I/C	The four functions in the statistical and substantive layers	Ch. 1; Arg. A/B	SUB-E situates, SUB-J augments and contextualizes, SUB-I interprets, SUB-C bounds the substantive reach of the corresponding statistical contribution
Argument diagrams	Support arrows and cross-layer links	Arrows show logical support within an argument; labelled links connect the meanings of the two layers	Ch. 1; Arg. A/B	Neither is a time sequence or causal path; label cross-question dependencies by the receiving function

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Numbers	Displayed values	Calculated results rounded once to four decimals with trailing zeros dropped (3.1, not 3.1000), and with no thousands separator; values shown as one set, such as a table column or an interval's limits, share the fewest decimals, up to four, that show every member; the leading zero is kept except for bounded quantities (p, r, R^2, proportions, probabilities: .0059); a nonzero value that rounds to zero is written as a bound ($p < .0001$), and a proportion or probability that rounds to 1 without equalling 1 keeps four decimals; negative numbers with a true minus sign (-0.2179) in prose, equations, and every authored table; counts, ordinary df, and codes as integers, and a count of 1,000 or more takes a comma in text and tables	Ch. 1	Native software output printed as output and figure axis labels keep their own format; that output and the text R draws in a figure keep R's hyphen before a negative value
GLM	π_i, $\pi(x)$, $\hat{\pi}_i$	Conditional event probability and its fitted value	Ch. 11; Ch. 13	Chapter 12 writes category probabilities as $P(y = k)$ (row below) and uses π_i for a zero-inflated model's probability that unit i belongs to the always-zero component

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
GLM	K, π_k; J, $P(y = k)$	Number of categories and the probability of category k	Ch. 2; Ch. 12	Chapter 2 writes the probabilities of one categorical variable's K categories as $\pi_1, \ldots, \pi_K$; Chapter 12 writes an outcome's J categories with conditional probabilities $P(y = k)$, category linear predictors η_k and ordinal thresholds ζ_k. In AIC and BIC, K counts the estimated parameters (Likelihood row)
GLM	η_i, logit	Linear predictor; the logit link $\text{logit}(\pi) = \log\{\pi/(1-\pi)\}$ and its inverse logit^{-1}	Ch. 11; Ch. 12	Both chapters write the conditional variance as $\phi\, V(\mu)$, with dispersion ϕ and variance function $V(\mu)$, and the general link as $g(\cdot)$
GLM	D, G^2	Deviance, and the likelihood-ratio statistic, referred to a chi-square distribution whose df count the parameters compared	Ch. 11; Ch. 12	Chapter 11 writes $D = -2\widehat{\ell}$ for ungrouped binary records and G^2 for the drop in deviance between nested models; Chapter 12 writes $D = 2\{\ell_{\text{saturated}} - \ell\}$, and for a table $G^2 = 2\sum O \log(O/E)$, a loglinear model's deviance against the saturated model, beside Pearson's X^2
Prediction	MAE	Mean absolute error of held-out predictions	Ch. 13	RMSE is the square root of the mean squared error.